

Engineering Trustworthy AI for Renewable-Powered Grids: A China-Informed New-Engineering Framework

English Research Report

September 2026

Scope. This report examines a concrete global issue: how to integrate rapidly expanding variable renewable electricity while artificial-intelligence (AI) workloads add a concentrated source of demand. It reads China's new-engineering reform and infrastructure practice from a Chinese standpoint, while testing which mechanisms can travel across institutional and income contexts.

Evidence status. Findings are based on a structured secondary-source review and comparative case analysis. Numerical targets in the solution section are explicitly ex-ante pilot targets, not claims about observed performance.

Abstract

The rapid build-out of wind and solar power is essential to climate mitigation, yet variable output, congested networks, cyber risk, and unequal technical capacity make integration difficult. At the same time, digital-sector loads (data centres, AI, and cryptocurrency) are becoming material and geographically concentrated. This report asks how a *New Engineering* approach can use AI without trading reliability, environmental integrity, security, or equity for speed. A structured desk review of international statistics, peer-reviewed research, Chinese policy, and two Chinese infrastructure cases was combined with a five-dimension coding framework. Four coupled gaps emerge: uncertainty is weakly connected to operational decisions; compute energy, water, and embodied carbon are under-counted; engineering, machine-learning, cyber, and public-interest capabilities remain separated; and data and standards are fragmented. China's new-engineering policies, renewable scale, Zhangbei flexible direct-current demonstration, and East-to-West Computing project illustrate coordination across curriculum, infrastructure, and public standards. The report proposes a four-layer **NE–AI Grid**: physics-informed probabilistic intelligence with deterministic fallbacks; practice-based interdisciplinary capability; carbon- and water-aware compute; and accountable, interoperable governance. A staged 60-month pilot adds independent evaluation, cost disclosure, and equity gates. The contribution is a China-informed but non-exclusive model that treats trustworthy AI as a socio-technical engineering capability rather than a software add-on.

Keywords: new engineering; artificial intelligence; renewable-energy integration; smart grids; China; trustworthy AI; energy equity; engineering education

Contents

1	Introduction	1
1.1	Background: an energy transition with a new load profile	1
1.2	Why a China-informed perspective matters	1
1.3	Problem statement, questions, and objectives	2
2	Research Design	2
2.1	Overall logic and analytical framework	2
2.2	Structured desk review	2
2.3	Coding, comparison, and credibility	3
2.4	Case selection	3
3	Results	4
3.1	Global pressure is simultaneous, not sequential	4
3.2	What the technical literature supports	5
3.3	Chinese policy and practice: coordination as an engineering capability	5
3.4	Four coupled gaps	6
3.5	Robustness checks on the diagnosis	6
4	A China-Informed Solution: the NE–AI Grid Framework	6
4.1	Design principles	6
4.2	Four layers and their interfaces	7
4.2.1	Layer 1: trustworthy technical intelligence	8
4.2.2	Layer 2: new-engineering capability	8
4.2.3	Layer 3: green compute	8
4.2.4	Layer 4: trust, standards, and global cooperation	8
4.3	Implementation roadmap and evaluation	8
4.4	Expected outcomes and theory of change	10
5	Discussion and Conclusion	10
5.1	Answering the research questions	10
5.2	Implications for China and the wider world	10
5.3	Limitations and improvement directions	11
5.4	Conclusion	11
	References	12

1 Introduction

1.1 Background: an energy transition with a new load profile

The electricity system is being rebuilt around weather-dependent resources. Renewable capacity reached about 3,870 GW at the end of 2023, with roughly 510 GW added that year (IRENA, 2024; IEA, 2024b). Wind and solar cut emissions, but variability raises the value of forecasting, flexibility, storage, transmission, and capable institutions (IPCC, 2023).

AI changes this problem in two ways. Machine-learning models can learn nonlinear links among weather, demand, network state, and equipment condition, improving short-horizon forecasts and maintenance. AI also belongs to a growing digital-sector load: the International Energy Agency (IEA) estimates that data centres, AI, and cryptocurrency used roughly 460 TWh in 2022 (nearly 2% of global demand) and could exceed 1,000 TWh by 2026 (IEA, 2024a). Without locational and temporal coordination, decarbonisation services can increase congestion, water stress, and fossil back-up generation.

The concrete global question is: *how can countries integrate variable renewables and AI-related loads while maintaining reliability, low lifecycle emissions, cybersecurity, and fair access?* It is also an education and policy question about who validates systems and bears risk.

1.2 Why a China-informed perspective matters

China is a consequential setting. It supplied over 50% of global renewable-capacity additions in 2023, and the IEA expects it to account for almost 60% of new renewable capacity becoming operational globally by 2028 (IEA, 2024b); renewable power capacity exceeds half of China's total installed power-generation capacity (National Energy Administration, 2024). This scale tests coordination among power electronics, dispatch, digital infrastructure, and workforce.

China's policy response links education and infrastructure. The Ministry of Education's 2017 response endorsed the university-led "Fudan Consensus" and "Tianda Action" and supported New Engineering projects built on interdisciplinary, industry-linked, outcome-based learning (Ministry of Education of the People's Republic of China, 2017; Ministry of Education, 2018a). National AI plans connect capability to "AI+X" education, while East-to-West Computing links digital infrastructure to regional resources (State Council of the People's Republic of China, 2017; Ministry of Education of the People's Republic of China, 2018; National Development and Reform Commission, 2022). These mechanisms are not templates: transfer depends on local governance, markets, data rights, and skills.

Within China's "Four New" agenda, the report focuses on New Engineering because renewable-grid AI risks must be engineered, taught, and governed together.

1.3 Problem statement, questions, and objectives

Smart-grid debates often isolate forecast error from the conditions that make deployment safe and durable; education debates celebrate interdisciplinarity without an operational test. We ask:

1. **RQ1:** What coupled technical, environmental, workforce, and governance bottlenecks limit trustworthy AI-enabled integration of variable renewable electricity?
2. **RQ2:** What lessons can be learned from China's new-engineering policy and renewable-grid/computing practices, and which lessons are transferable rather than context-bound?
3. **RQ3:** What implementable China-informed framework, pilot sequence, and evaluation metrics can align reliability, decarbonisation, security, and equity?

The objectives are to diagnose the coupled problem, connect Chinese practice to global standards, and specify a falsifiable solution with accountable gates. The contribution is a triangulated socio-technical analysis and pilot-ready curriculum–infrastructure–governance package.

2 Research Design

2.1 Overall logic and analytical framework

The study uses a structured desk review with qualitative comparative case analysis; it contains no primary data or causal estimation. Figure 1 traces the chain from global pressure to coded evidence, Chinese cases, and safety and equity gates. The unit of analysis is a *deployment arrangement*: algorithms, data, assets, people, rules, and affected communities.

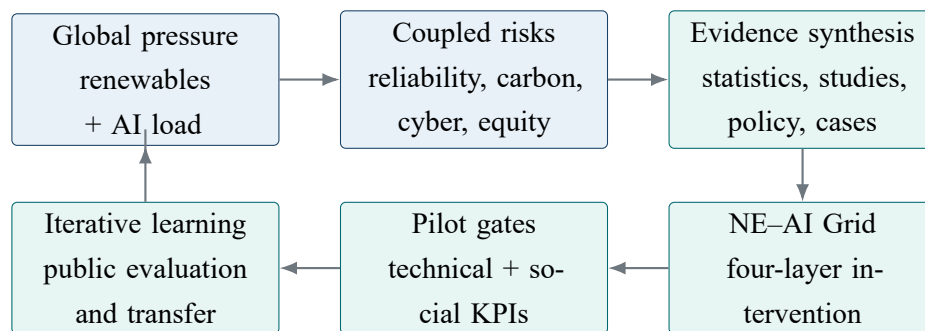


Figure 1. Study logic: from a global deployment problem to an evaluated and transferable intervention.

Source: Author's synthesis of the research design.

2.2 Structured desk review

The review covered English- and Chinese-language materials from January 2017 through December 2024. Searches used Google Scholar, journal indexes, IEA, IRENA, IPCC, ITU, NIST, UNESCO, and Chinese government and utility sites. The fixed string was (renewable OR wind

OR solar) AND (grid OR dispatch OR curtailment) AND (AI OR machine learning OR forecast) AND (education OR skills OR governance). Citation chaining followed high-relevance reviews; a dated export records source, year, geography, hit, and disposition.

Items were included when they addressed integration or an enabling capability, reported an inspectable method, statistic, policy instrument, or project outcome, and identified geography and period. Opinion pieces without traceable evidence, vendor marketing without methods, and duplicates were excluded. Counts mark this evidence boundary, not the literature's size.

Table 1. Documented screening and extraction log

Stage	Records	Rule or action
Searches and exports	126	Title, year, source, geography, and link recorded.
Deduplication and metadata check	89	37 duplicate or inaccessible records removed.
Title/abstract screen	54	Explicit grid, AI, education, or governance link.
Full-text eligibility	38	Inclusion and traceability checks applied.
Synthesis set	31	18 peer-reviewed, 9 institutional, and 4 Chinese policy/project items.

Source: Author-compiled screening log; counts describe this desk-review process, not the total literature.

2.3 Coding, comparison, and credibility

Each item was coded on five dimensions: forecast/dispatch; reliability/curtailment; compute footprint; workforce/curriculum; and governance, cybersecurity, and equity. Metrics stayed in original units because horizons, sites, baselines, and curtailment definitions differ. Statistics required a denominator and year; project claims were labelled “project-reported” unless independently evaluated; algorithmic gains were conditional when site or data dependence was reported. A single-reviewer recheck tested consistency, not inter-rater reliability. No personal data were collected.

2.4 Case selection

Two purposively selected Chinese cases expose different coordination problems: Zhangbei links wind, solar, storage, and long-distance DC delivery, while East-to-West Computing links data-centre hubs to regional power and land conditions. We compared them with NIST, IEC 62443, ISO/IEC 42001, and UNESCO/OECD principles (NIST, 2023; IEC, 2018; ISO, 2023; UNESCO, 2021; OECD, 2019). They are illustrative, non-causal cases; comparison asks what

mechanism is observable and what adaptation a lower-capacity setting requires.

3 Results

3.1 Global pressure is simultaneous, not sequential

Table 2 combines indicators that are often discussed separately. The first four quantify deployment scale and concentration; the next two capture digital demand; the final two indicate capability constraints.

Table 2. Selected indicators defining the global deployment problem

Indicator	Reported value	Interpretation for new engineering and AI
Global renewable capacity, end-2023	3,870 GW	System-level interoperability and flexibility.
Global renewable additions in 2023	about 510 GW	Deployment can outpace training and commissioning.
China's share of 2023 additions	over 50%	Global learning value; test scale assumptions.
China's expected share of global new capacity operational by 2028	almost 60%	Chinese choices affect the near-term pipeline.
Digital-sector electricity (data centres, AI, cryptocurrency), 2022	about 460 TWh (about 2%)	Location, timing, and carbon intensity matter.
Projected digital-sector electricity, 2026	more than 1,000 TWh	Compute growth is uncertain demand.
People offline, 2023	2.6 billion	Fund access, training, and offline fallbacks.
Employers expecting workers' skills to be disrupted	44% (2023 survey estimate)	Reskilling is a design requirement.

Source: IEA (2024a, 2024b), IRENA (2024), ITU (2023), and WEF (2023); source precision retained.

Together, the indicators show a feedback loop: renewable growth increases the value of forecasting and flexibility, while schedulable digital demand can support or strain the grid. Compute consumes water and concentrates benefits among organisations with data and skilled staff. Capability and rebound effects are therefore deployment constraints (International Telecommunication Union, 2023; World Economic Forum, 2023; Kaack et al., 2022; Strubell et al., 2019; Rolnick et al., 2022).

3.2 What the technical literature supports

The forecasting literature supports a direction, not a universal magnitude: deep-learning and ensemble methods often reduce short-term error against persistence or single-model baselines (Wang et al., 2019; Hong et al., 2020), but gains vary with horizon, weather, data, and market rules. Distribution shift and sensor faults remain failure modes. Physics-informed learning can impose conservation, power-flow, and equipment constraints (Karniadakis et al., 2021). In our screened set, reliability, lifecycle carbon, water, cyber incidents, and distributional effects were seldom reported together, encouraging narrow optimisation.

3.3 Chinese policy and practice: coordination as an engineering capability

Table 3. Chinese policy and infrastructure milestones relevant to NE–AI Grid

Year	Milestone	Observable mechanism and relevance
2017	Fudan Consensus and Tianda Action	Interdisciplinary, practice-oriented new engineering.
2017	New Generation AI Development Plan	National direction for AI, talent, and governance.
2018	MoE first-batch New Engineering projects	Industry–university projects and outcome-based curricula.
2022	East-to-West Computing project	Eight hubs and ten clusters link networks, land, and energy.
2020–present	Zhangbei flexible DC demonstration and related projects	Wind, solar, storage, power electronics, and delivery; AI metrics limited.
2023–2024	Renewable capacity exceeds half of China’s installed power capacity	Flexibility, forecasting, and commissioning become national priorities.

Source: MoE (2017, 2018a), State Council (2017), NDRC (2022), Hitachi Energy (2020), China Daily (2020), and NEA (2024).

Zhangbei: physical co-design. Commissioned in 2020, Zhangbei is a four-station meshed voltage-source-converter HVDC grid linking wind, solar, storage, and Beijing delivery. Profiles report 500 kV, up to 4,500 MW, and 14 billion kWh annual clean delivery (Hitachi Energy, 2020; China Daily, 2020). Its mechanism is shared interfaces and tests across engineering and regulatory teams. Public sources lack audited AI, curtailment, or lifecycle series; this is evidence of co-design, not causal AI performance.

East-to-West Computing: digital and energy co-location. The 2022 launch describes eight hubs and ten clusters linking demand, networks, and regional resources (National Development and Reform Commission, 2022). Renewable-rich siting may improve clean-power use but shift water or community impacts. Appraise compute placement with the same carbon, water, reliability, and equity accounts as generation; lower-capacity settings may need cooperatives or development finance.

3.4 Four coupled gaps

Coding and case comparison produced the diagnosis in Table 4. The gaps reinforce one another; addressing one alone leaves deployment fragile.

Table 4. Diagnosis of coupled deployment gaps

Gap	Evidence pattern	Consequence if untreated
Uncertainty-to-action gap	Forecast uncertainty rarely reaches reserve, storage, or protection decisions.	Overconfidence can increase imbalance, curtailment, or overrides.
AI footprint gap	Appraisals omit model energy, water, hardware, and embodied carbon.	Low-carbon applications can create peaks, stress, or rebound emissions.
Siloed capability gap	Power, ML, cyber, ethics, and community skills remain separate.	Teams struggle to maintain a safe system.
Fragmented trust and access gap	Ownership, standards, incidents, and connectivity vary across settings.	Models become locked-in and hard to adapt.

Source: Author's synthesis of the included evidence set.

3.5 Robustness checks on the diagnosis

An exploratory qualitative check contrasted records with and without policy documents and higher- and lower-income references. Higher-income records emphasised accuracy and cyber controls; lower-income records emphasised connectivity, finance, and technician shortages. Neither Chinese case proves that coordination lowers emissions or improves reliability, so the pilot pre-registers baselines and reports null or adverse outcomes.

4 A China-Informed Solution: the NE–AI Grid Framework

4.1 Design principles

The framework turns the diagnosis into four operating principles:

1. **Safe uncertainty:** enforce physical constraints, expose calibrated intervals, and retain a deterministic, human-authorised fallback.
2. **Accounted capability:** appraise compute energy, water, hardware, and carbon, and fund curricula, apprenticeships, and maintenance.
3. **Open accountability:** require auditability, red-teaming, portable interfaces, and incident disclosure.
4. **Local adaptation:** share Chinese mechanisms and benchmarks; local actors choose institutions and benefits.

4.2 Four layers and their interfaces

Figure 2 presents four mutually dependent layers. Their interfaces are acceptance conditions: a model needs a data and safety specification, a curriculum is assessed on a live system, and compute procurement reports environmental and social indicators.

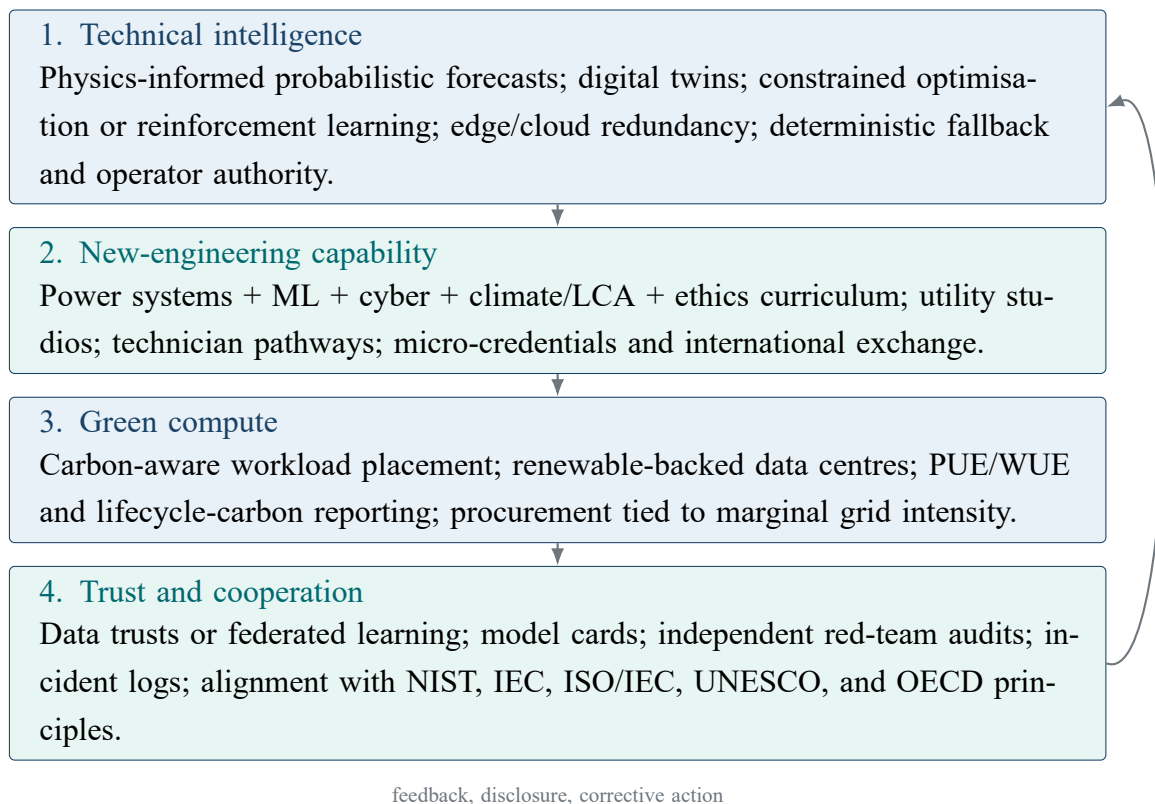


Figure 2. The NE–AI Grid architecture. Each layer has an operational deliverable and a verification interface.

Source: Author’s framework, informed by Karniadakis et al. (2021), NIST (2023), and Chinese new-engineering policy.

4.2.1 Layer 1: trustworthy technical intelligence

Layer 1 combines a physical baseline with probabilistic learning. A day-ahead model outputs $p(y_{t+h} \mid x_t)$ and calibrated intervals; a digital twin checks power-flow, ramp, voltage, and protection limits. Constrained dispatch treats reliability as a hard constraint and cost/emissions as objectives. Operators retain authority; low confidence or lost connectivity triggers a tested deterministic policy. Required artefacts are a sensor sheet, model card, seasonal hold-outs, threat model, incident playbook, and versioned API. Federated learning does not replace access control or validation.

4.2.2 Layer 2: new-engineering capability

Layer 2 organises a postgraduate track around a live grid problem: power markets and dynamics, statistical learning, digital twins, cyber safety, lifecycle assessment, ethics, and communication. Utility–university studios use anonymised data and hardware-in-the-loop tests; assessment rewards a safe operating envelope and reproducibility. Technicians, operators, procurement officers, and regulators receive stackable credentials. Chinese partners share benchmarks and faculty exchange while host institutions retain control.

4.2.3 Layer 3: green compute

Layer 3 reports energy per training run and defined inference, WUE, PUE, hardware replacement, and marginal grid carbon. A carbon-aware scheduler moves deferrable workloads subject to latency, sovereignty, and reliability limits. Clean-power co-location requires published water and community safeguards.

4.2.4 Layer 4: trust, standards, and global cooperation

Layer 4 assigns stewardship, access tiers, retention limits, and benefit-sharing through a data trust or cooperative. Redacted model cards and incident logs are public; high-impact changes trigger independent review and red-teaming. NIST, IEC 62443, ISO/IEC 42001, and UNESCO/OECD principles supply compatible vocabularies (NIST, 2023; IEC, 2018; ISO, 2023; UNESCO, 2021; OECD, 2019). Lower-income regions need concessional finance, offline tools, and train-the-trainer partnerships.

4.3 Implementation roadmap and evaluation

The proposal is a staged learning programme, not a one-time procurement. Table 5 gives ex-ante targets, not observed results. Mean absolute error (MAE) measures forecast accuracy; SAIDI/SAIFI measure interruption duration/frequency; PUE, WUE, and lifecycle assessment (LCA) capture compute impacts. Each gate requires a costed business case for capital, operations, compute, training, and decommissioning; scale-up cannot exceed a regulator-approved total-cost-of-ownership ceiling.

Table 5. Five-year pilot roadmap and ex-ante targets

Window	Actions	Decision gate and illustrative target
0–12 months (baseline)	Select a renewable-heavy region; map data, assets, communities, and cyber dependencies; establish a utility–university studio; publish operational and compute baselines.	No automation until data, safety, consultation, and cost checks pass; version 100% of inputs.
12–36 months (shadow and limited pilot)	Run probabilistic forecasts and dispatch in shadow mode, then on a bounded feeder; use seasonal hold-outs, hardware-in-the-loop tests, and an independent red team.	Test at least 15% lower MAE and 10% lower curtailment; SAIDI/SAIFI non-inferior within a pre-specified margin and costs within the ceiling.
36–60 months (scale and transfer)	Connect utilities or provinces; add carbon-aware scheduling and federated learning; release a benchmark and train instructors in a lower-income partner region.	Test at least 20% lower carbon per defined inference, 100% auditable high-impact actions, and distributional effects; scale after two seasonal evaluations.

Source: Targets are proposed ex-ante pilot thresholds. Actual values must be estimated with confidence intervals, pre-registered baselines, and an independent evaluator.

The utility owns safety and baseline KPIs; the university owns curriculum and benchmark artefacts; procurement reports lifecycle cost; and a utility–university–regulator–worker–community committee can pause deployment or hear appeals. Evaluation reports point estimates, confidence intervals, prediction-interval coverage, subgroup effects, and incidents against a pre-registered baseline, with weather and market conditions recorded. Scale-up is barred when outage risk, water stress, lifecycle cost, or exclusion exceeds its margin. Paid apprenticeship seats and offline access are equity outputs.

4.4 Expected outcomes and theory of change

Near-term, operators receive calibrated information before automatic action. Medium-term outcomes are lower curtailment, better storage/transmission use, and a smaller compute footprint; long-term outcomes are portable competence, open benchmarks, and adaptable governance. The causal chain is conditional: data and training enable validation, validation plus flexibility enables gains, and accounting plus participation determine legitimacy.

5 Discussion and Conclusion

5.1 Answering the research questions

For **RQ1**, the principal barrier is coupling, not a lack of algorithms: uncertainty, physical constraints, compute externalities, fragmented skills, and unequal access interact. A point-forecast benchmark cannot show whether deployment improves reliability or shifts emissions and risk.

For **RQ2**, China's mechanism-level lesson is coordination. New-engineering policy incentivises boundary crossing; Zhangbei shows why physical projects need it; and East-to-West Computing makes compute location and timing a planning variable. Administrative scale, procurement, and finance are context-bound. Transfer should share interfaces, tests, curriculum outcomes, and accounting rules while leaving ownership and participation local; these are pilot hypotheses, not causal findings.

For **RQ3**, NE-AI Grid is feasible because it starts bounded, keeps human authority, and ties scale to evidence. Its novelty is to place education, green-compute accounting, and accountable governance in the same acceptance test as forecast and dispatch.

5.2 Implications for China and the wider world

For China, the framework adds public evaluation and lifecycle disclosure to coordinated infrastructure. Programmes could require interoperable APIs, model cards, and auditable environmental metrics, while universities assess competence on operational problems. International collaboration can focus on open benchmarks, train-the-trainer studios, and finance for local data stewardship. The approach is designed to be consistent with NIST, IEC, ISO/IEC, UNESCO, and OECD principles and can operate through regional or municipal utilities.

Trade-offs are explicit: carbon-aware scheduling can conflict with latency or data sovereignty; federated learning can complicate debugging; and automation can displace routine work before retraining. Each trade-off receives an owner and review date. Community representatives receive plain-language risk, water, and benefit summaries and a channel to contest siting or service changes, making equity auditable.

5.3 Limitations and improvement directions

The study relies on public secondary sources that favour visible successes; Chinese project data and AI metrics are not always internationally comparable. Screening counts are a process record, not a probability sample; heterogeneous definitions prevent a pooled effect; and the two cases cannot establish global transfer. KPI thresholds are hypotheses requiring pre-registration, confidence intervals, and independent evaluation. Future work should build a privacy-preserving cross-climate benchmark, test curtailment and reliability longitudinally, quantify water and embodied carbon, and study worker and community outcomes under different ownership and data regimes.

5.4 Conclusion

Renewable build-out and rising AI loads make grid integration a shared engineering challenge. China's experience suggests that scale matters when education, infrastructure, and standards are coordinated. NE–AI Grid turns that mechanism into a testable programme: physics-aware intelligence with a fallback, interdisciplinary project-based capability, explicitly accounted compute impacts, and governance that makes data, failures, and benefits visible. It offers a route toward a reliable, low-carbon, and more equitable digital energy transition.

References

- International Electrotechnical Commission. (2018). *IEC 62443: Industrial communication networks—Network and system security*. Geneva: IEC. <https://www.iec.ch/industrial-cybersecurity>.
- Hong, T., Pinson, P., Wang, Y., Weron, R., Yang, S., & Zareipour, H. (2020). Energy forecasting: A review and outlook. *IEEE Open Journal of Power and Energy*, 7, 376–388. <https://doi.org/10.1109/OAJPE.2020.3029979>.
- International Energy Agency. (2024a). *Electricity 2024: Analysis and forecast to 2026*. Paris: IEA. <https://www.iea.org/reports/electricity-2024>.
- International Energy Agency. (2024b). *Renewables 2023: Analysis and forecast to 2028*. Paris: IEA. <https://www.iea.org/reports/renewables-2023>.
- Intergovernmental Panel on Climate Change. (2023). *Climate change 2023: Synthesis report*. Geneva: IPCC. <https://doi.org/10.59327/IPCC/AR6-9789291691647>.
- International Telecommunication Union. (2023). *Facts and figures 2023*. Geneva: ITU. <https://www.itu.int/itu-d/reports/statistics/facts-figures-2023/>.
- International Renewable Energy Agency. (2024). *Renewable capacity statistics 2024*. Abu Dhabi: IRENA. <https://www.irena.org/Publications/2024/Mar/Renewable-capacity-statistics-2024>.
- Kaack, L. H., Donti, P. L., Strubell, E., Kamiya, G., Creutzig, F., & Rolnick, D. (2022). Aligning artificial intelligence with climate change mitigation. *Nature Climate Change*, 12, 518–527. <https://doi.org/10.1038/s41558-022-01377-7>.
- Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, 3, 422–440. <https://doi.org/10.1038/s42254-021-00314-5>.
- Ministry of Education of the People's Republic of China. (2017). *Response on promoting emerging-engineering education reform* [policy response]. Beijing: MoE. http://www.moe.gov.cn/jyb_xxgk/xxgk_jyta/jyta_gaojiaosi/201712/t20171219_321828.html.
- Ministry of Education of the People's Republic of China. (2018a). *Notice on the first batch of new engineering research and practice projects* [policy document]. Beijing: MoE. http://www.moe.gov.cn/srcsite/A08/s7056/201803/t20180329_331767.html.
- Ministry of Education of the People's Republic of China. (2018). *AI Innovation Action Plan for Higher Education* [policy document]. Beijing: MoE. http://www.moe.gov.cn/srcsite/A16/s7062/201804/t20180410_332722.html.

- National Development and Reform Commission. (2022). *East-to-West Computing officially launches: Eight hubs energise the digital economy* [policy briefing]. Beijing: NDRC. https://www.ndrc.gov.cn/fzggw/jgsj/gjss/sjdt/202203/t20220321_1319862.html.
- National Energy Administration of the People's Republic of China. (2024). *2023 national electricity industry statistics*. Beijing: NEA. https://english.www.gov.cn/news/202401/13/content_WS65a22a99c6d0868f4e8e30aa.html.
- National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AIRMF 1.0)* (NIST AI 100-1). Gaithersburg, MD: NIST. <https://doi.org/10.6028/NIST.AI.100-1>.
- Organisation for Economic Co-operation and Development. (2019). *Recommendation of the Council on Artificial Intelligence* (OECD/LEGAL/0449). Paris: OECD. <https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449>.
- International Organization for Standardization. (2023). *ISO/IEC 42001:2023 Information technology—Artificial intelligence—Management system*. Geneva: ISO. <https://www.iso.org/standard/81230.html>.
- Rolnick, D., Donti, P. L., Kaack, L. H., Kochanski, K., Lacoste, A., Sankaran, K., ... Bengio, Y. (2022). Tackling climate change with machine learning. *ACM Computing Surveys*, 55(2), Article 42. <https://doi.org/10.1145/3485128>.
- State Council of the People's Republic of China. (2017). *New Generation Artificial Intelligence Development Plan* [policy document]. Beijing. http://www.gov.cn/zhengce/content/2017-07/20/content_5211996.htm.
- Hitachi Energy. (2020). *Zhangbei: The world's first DC-grid with HVDC Light technology* [technical project profile]. <https://www.hitachienergy.com/news-and-events/customer-stories/zhangbei>.
- China Daily. (2020, July 1). World's first flexible DC power grid starts operation. <https://www.chinadaily.com.cn/a/202007/01/WS5efc061ea3108348172566e2.html>.
- Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 3645–3650). <https://doi.org/10.18653/v1/P19-1355>.
- United Nations Educational, Scientific and Cultural Organization. (2021). *Recommendation on the ethics of artificial intelligence*. Paris: UNESCO. <https://unesdoc.unesco.org/ark:/48223/pf0000381137>.

Wang, H., Lei, Z., Zhang, X., Zhou, B., & Peng, J. (2019). A review of deep learning for renewable energy forecasting. *Energy Conversion and Management*, 198, 111799. <https://doi.org/10.1016/j.enconman.2019.111799>.

World Economic Forum. (2023). *The future of jobs report 2023*. Geneva: WEF. <https://www.weforum.org/reports/the-future-of-jobs-report-2023/>.